Airline Reservation System Analytics

A reproducible SQL case study using intentionally synthetic data. Author: JahMari Harrigan, with Codex development assistance.

Question and analytical approach

How do reservation outcomes and booked value vary across routes and booking months, and are the underlying records internally consistent? The analysis begins with the normalized operational schema, distinguishes bookings from passengers, checks data quality, and aggregates at the appropriate grain. SQL queries include CTEs, grouped joins and DENSE_RANK.

Measure	Reproduced result
Reservations	660
Confirmed	528
Cancelled	132
Cancellation rate	20.0%
Net booked value	$105,080
June and August net value	$24,620 each
July and September net value	$27,920 each

Interpretation

July and September have $3,300 more net booked value than June and August, despite identical monthly booking counts and cancellation counts. In this scenario, value differs through the generated fare and route mix. This is a demonstration of metric reasoning, not evidence of real seasonality. A real business study would need observed demand, costs, market context and uncertainty analysis.

Reproduce and extend

Create a fresh database, run setup and seed-analytics.js, then execute analytics/analysis.sql and compare the administrator dashboard. Read METRICS.md for definitions and exclusions. Keep capacity at the departure grain to avoid counting seats once per booking. Before drawing conclusions, verify the exception report returns no rows.